# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Scenario-Based Research Assessment

**COURSE NAME:** **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

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**COURSE OUTCOME COVERED**

**CO1:** *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

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**Total Marks: 50** **Time: 60 Mins**

**No. of Questions: 5**

Annexure A

**Scenario-Based Research Assessment**

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***Code of Conduct:***

*I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

| Criterion | Weightage | Marks Obtained |
|---|---|---|
| **Identification of Latest AI Application** | 10% | |
| **Problem Analysis and Domain Understanding** | 15% | |
| **AI Technologies and Working Mechanism** | 20% | |
| **Research Quality and Literature Review** | 10% | |
| **Real-World Implementation and Case Analysis** | 10% | |
| **Impact, Challenges, and Future Scope** | 15% | |
| **Originality and Critical Thinking** | 10% | |
| **Organization, Presentation, and Documentation** | 10% | |
| **Total** | 100% | |

Signature of the Faculty Total Marks Awarded

## Abstract

Artificial Intelligence (AI) is rapidly transforming the delivery of healthcare across diagnosis, treatment planning, and patient monitoring. Among the many domains of clinical AI, medical imaging and radiology have emerged as the most mature and widely deployed application area, accounting for the majority of AI-based devices authorised by regulatory bodies such as the United States Food and Drug Administration (FDA). This report examines AI-based radiology systems as a representative and current example of AI adoption in healthcare. It explains the underlying working principles of these systems, the specific AI and machine learning techniques employed — including convolutional neural networks, deep learning segmentation models, and emerging multimodal foundation models — and documents real-world deployments such as Aidoc's aiOS platform and Viz.ai's stroke-detection network, both of which now operate across thousands of hospitals worldwide. The report further analyses the benefits these systems bring in terms of diagnostic speed, triage prioritisation, and clinician workload reduction, alongside the technical, regulatory, and ethical limitations that continue to shape their adoption. The findings indicate that while AI radiology tools have achieved significant clinical traction, especially in time-critical conditions such as stroke and pulmonary embolism, challenges relating to workflow integration, algorithmic bias, explainability, and reimbursement remain unresolved and will define the next phase of AI adoption in clinical practice.


## Introduction

Healthcare systems worldwide face mounting pressure from ageing populations, a rising burden of chronic disease, and a persistent shortage of specialist clinicians. At the same time, the volume of diagnostic data generated — particularly medical images from computed tomography (CT), magnetic resonance imaging (MRI), and X-ray modalities — has grown far faster than the workforce available to interpret it. Artificial Intelligence, and specifically deep learning, has stepped into this gap as a tool capable of analysing large volumes of complex clinical data with a level of consistency and speed that human interpretation alone cannot match.

AI in healthcare spans a wide spectrum of applications, ranging from AI-assisted diagnosis and radiology, to robotic-assisted surgery, computational drug discovery, and conversational virtual health assistants. Among these, AI-based radiology and diagnostic imaging systems represent the most clinically embedded and commercially deployed category today. This is reflected in regulatory data: as of early 2026, radiology accounts for approximately 76 percent of all AI-enabled medical devices cleared by the FDA, far outweighing every other clinical specialty combined. This report focuses on this category as the chosen latest AI-based healthcare application, tracing its technical foundations, real-world rollout, and the benefits and limitations that accompany its use.

The objectives of this report are fourfold: first, to explain the working principles by which AI radiology systems convert a raw medical image into an actionable clinical alert; second, to identify the specific families of AI techniques, from convolutional neural networks to emerging foundation models, that underpin these systems; third, to document verified, real-world implementations at named health systems and vendors as evidence of practical adoption rather than theoretical capability; and fourth, to critically evaluate the benefits and limitations that determine how, and how quickly, this technology is likely to be adopted more broadly across global healthcare systems, including in resource-constrained settings.

The clinical cost of the underlying problem is well quantified in the literature and helps explain why radiology, rather than another specialty, became the focal point of clinical AI investment. In stroke care specifically, delayed identification of a large-vessel occlusion directly worsens functional outcomes, since each additional minute without treatment corresponds to further irreversible neuronal loss, a relationship widely summarised in stroke medicine as "time is brain." A 2025 narrative review of AI applications in stroke care confirmed that automated CT and MRI analysis can meaningfully shorten the interval between imaging and treatment decision, directly targeting this time-sensitive bottleneck rather than merely automating an administrative task. This framing clarifies that the problem AI radiology systems address is not simply a shortage of radiologist labour, but a structural mismatch between the speed at which certain conditions must be identified and the speed at which a traditional, sequential human reading and communication workflow can operate.

## Artificial Intelligence in Healthcare — An Overview

Before examining radiology AI specifically, it is useful to place it within the broader landscape of AI applications in healthcare, which can be broadly grouped into the following categories:

- AI-Assisted Medical Diagnosis: Machine learning models trained on patient records, laboratory results, and clinical notes to support diagnostic decision-making across specialties.
- AI Radiology and Imaging Systems: Deep learning models that detect, segment, and prioritise abnormalities in CT, MRI, X-ray, and ultrasound images.
- Robotic-Assisted Surgery: AI-enhanced surgical platforms that provide real-time guidance, tremor filtration, and predictive assistance during operative procedures.
- AI-Driven Drug Discovery: Generative and predictive models used to identify candidate molecules, predict protein structures, and accelerate clinical trial design.
- Virtual Healthcare Assistants: Conversational AI agents that support patient triage, appointment scheduling, medication reminders, and remote monitoring.

Of these categories, radiology AI was selected for detailed study in this report because it is presently the most widely deployed, most heavily regulated, and most extensively documented category of clinical AI, offering rich, verifiable, real-world evidence of both benefit and limitation.

## Selected Application: AI-Based Radiology and Diagnostic Imaging Systems

AI-based radiology systems are software platforms that use deep learning algorithms to automatically analyse medical images and flag findings that require urgent clinical attention. Rather than replacing the radiologist, these systems function primarily as triage and decision-support tools: they run in the background of a hospital's imaging workflow, analyse scans as soon as they are acquired, and alert the appropriate care team when a time-critical abnormality — such as a large-vessel occlusion stroke, pulmonary embolism, or intracranial haemorrhage — is detected.

Two vendors illustrate the scale this category has reached. Aidoc, whose aiOS platform now supports clinical decisions across nearly two thousand hospitals worldwide and has analysed over 120 million patient scans, and Viz.ai, whose stroke-detection network is deployed across more than 1,700 hospitals with over 50 FDA-cleared algorithms, together represent the clearest evidence that AI radiology has moved from research prototype to standard-of-care infrastructure in leading health systems.

This category was chosen for detailed study over the other candidate applications listed earlier, such as robotic surgery or drug discovery, for three reasons. First, its regulatory footprint is the largest and most transparent of any clinical AI category, since the FDA publishes a continuously updated public list of every cleared device, allowing adoption trends to be tracked with precision. Second, its clinical impact is measurable in concrete, time-based outcomes, such as minutes saved in stroke diagnosis, rather than in harder-to-verify claims. Third, it is presently the category with the largest gap between regulatory approval and everyday clinical use, making it an instructive case study in the practical barriers that separate a validated algorithm from a genuinely adopted clinical tool.

### Why Radiology Was an Early Adopter of AI

Radiology was uniquely positioned to adopt deep learning earlier than most other medical specialties for a combination of technical and structural reasons. Medical images are already stored in a standardised digital format (DICOM), which meant large, structured datasets were available for model training long before comparable datasets existed for unstructured clinical text or physical examination findings. In addition, the interpretive task performed by a radiologist, identifying and localising a visual pattern within an image, maps closely onto the pattern-recognition strengths of convolutional neural networks, which were originally developed for general computer vision tasks such as object detection. Finally, the rising volume of imaging studies, driven by an ageing population and expanding indications for CT and MRI, created sustained commercial and clinical demand for tools that could help manage caseload without a proportional increase in radiologist headcount.

## What the Imaging Machine Looks Like

Before examining how AI analyses a scan, it is useful to see the physical hardware that produces it. The illustration below shows a typical CT/MRI scanning system as it appears in a hospital imaging suite: a large ring-shaped gantry housing the scanning hardware, a motorised patient table that moves the patient into the circular bore at the centre of the gantry, and an adjoining operator console from which the technologist controls the scan and views the resulting images.

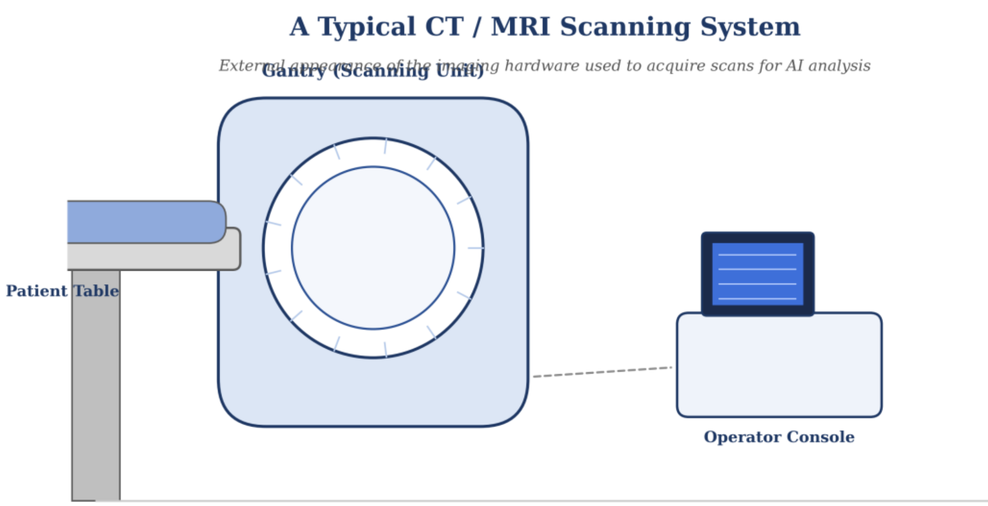


*Figure 1. External appearance of a typical CT/MRI scanning system, showing the gantry, patient table, and operator console.*

## Inside the Machine: Parts and Their Connection to AI

Beneath its outer housing, the gantry contains several core components that work together to produce a diagnostic image. The X-ray tube generates the radiation beam (in CT) or, in MRI, is replaced by strong superconducting magnets and radiofrequency coils; the detector array, arranged in a ring around the patient bore, captures the signal that passes through or is emitted by the body; and a dedicated reconstruction computer converts this raw signal into a viewable cross-sectional image. The labelled diagram below identifies these components and shows how the resulting image output feeds directly into the AI pipeline described in the following section, from AI inference and triage, through to the radiologist's workstation.

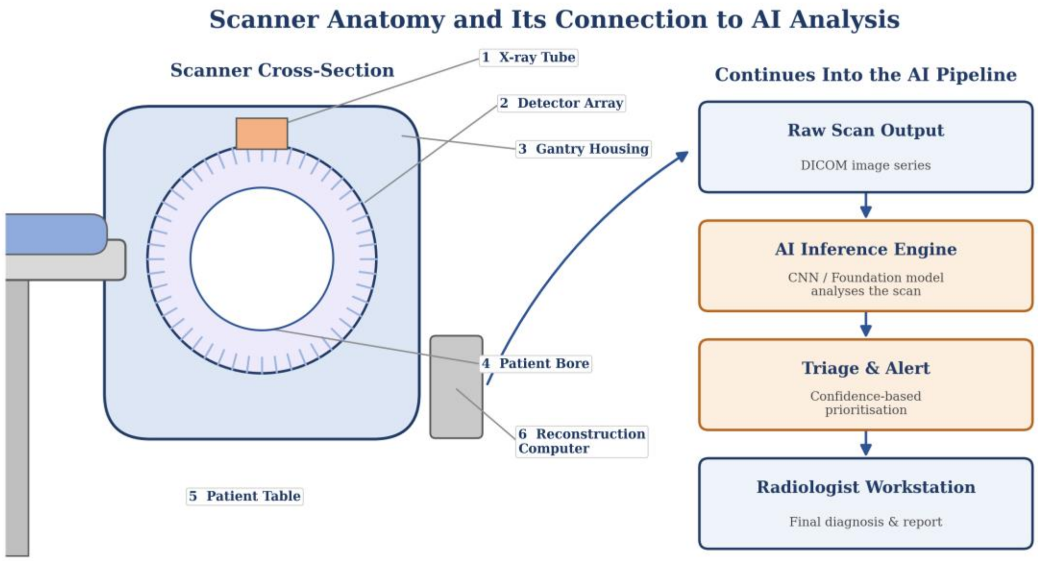


*Figure 2. Core components of the scanner (X-ray tube, detector array, gantry housing, patient bore, table, and reconstruction computer) and how the raw scan output feeds into the AI inference and radiologist review pipeline.*

## Working Principles

AI radiology systems operate through a structured pipeline that integrates directly with a hospital's existing imaging infrastructure, most commonly its Picture Archiving and Communication System (PACS). The working principle can be described in the following stages:

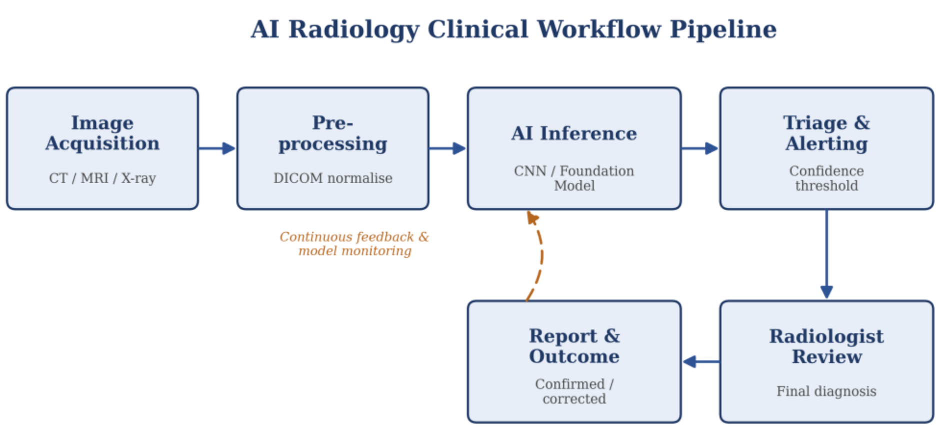


*Figure 3. End-to-end AI radiology clinical workflow, from image acquisition to radiologist review and continuous model monitoring.*

- Image Acquisition: A CT, MRI, or X-ray scan is acquired as part of routine clinical care and is automatically transmitted to the AI platform in parallel with its transmission to the radiologist's reading list.
- Pre-processing: The raw imaging data (typically in DICOM format) is normalised, resized, and denoised to a standard input format suitable for the underlying neural network.

- Feature Extraction and Inference: A trained deep learning model — typically a convolutional neural network (CNN) or a 3D volumetric variant — scans the image for patterns associated with specific pathologies, producing a probability score or a segmented region of interest.
- Triage and Alerting: If the model's confidence exceeds a validated clinical threshold, the platform sends an immediate alert to the relevant specialist, often directly to a smartphone application, along with the relevant image series.
- Radiologist Review: The flagged case is reprioritised to the top of the radiologist's worklist. The radiologist retains full responsibility for the final diagnosis; the AI output is a decision-support aid, not an autonomous diagnosis.
- Continuous Feedback: Outcomes and radiologist corrections are logged and, in many deployments, used to monitor model performance and inform periodic retraining.

This workflow is deliberately designed around the principle of triage rather than replacement: the AI system's primary clinical value lies in re-ordering the queue of cases so that the most time-critical ones reach a human expert first, rather than in issuing an independent diagnosis.

## Model Training and Validation Cycle

Behind every deployed model lies a training and validation cycle that begins well before the software reaches a hospital. Vendors assemble large, expert-labelled datasets of historical scans, often drawn from multiple partner health systems to capture variation in scanner hardware, patient demographics, and disease presentation. The model is trained to minimise the difference between its predictions and the expert-assigned ground truth labels, and its performance is then measured on a separate, held-out test dataset that it has never seen during training. Only once a model demonstrates consistent sensitivity and specificity on this held-out set, and subsequently on prospective clinical data, does it proceed toward regulatory submission. After deployment, most platforms continue to monitor live performance against radiologist-confirmed outcomes, a process sometimes referred to as post-market surveillance, to detect any gradual decline in accuracy, known as model drift, that might occur as patient populations or imaging equipment change over time.

## AI Techniques and Technologies Used

### Convolutional Neural Networks (CNNs)

Convolutional Neural Networks remain the foundational architecture for most FDA-cleared radiology AI tools. CNNs use layered filters to detect increasingly abstract visual features — from simple edges and textures in early layers to complex anatomical patterns in deeper layers — making them highly effective for classifying and localising abnormalities in two- and three-dimensional medical images.

### Image Segmentation Networks

Architectures such as U-Net and its 3D variants are used to perform pixel-level or voxel-level segmentation, for example to precisely outline the boundary of a haemorrhage or tumour. This is essential for tools such as Viz Subdural Plus, which automatically quantifies the volume and thickness of subdural haemorrhages rather than simply flagging their presence.

### Foundation Models and Multimodal AI

A significant recent development is the emergence of foundation models trained on very large, diverse imaging datasets that can be adapted to multiple downstream tasks. Aidoc's CARE foundation model, which received FDA clearance in January 2026 as the first multi-condition triage solution for body CT, exemplifies this shift: a single model is able to detect and prioritise up to fourteen distinct conditions from one scan, rather than requiring a separate narrow model for each pathology.

### Natural Language Processing (NLP) and Generative AI

A newer frontier involves using large language model-style architectures to draft preliminary radiology reports from image findings. Aidoc's First Read tool, granted FDA Breakthrough Device Designation in June 2026, analyses chest radiographs and generates draft report text for radiologist review, aiming to reduce the growing backlog in outpatient imaging turnaround time.

### Predictive and Risk-Stratification Models

Beyond acute triage, models such as Mirai are used for longer-horizon risk prediction — for instance, estimating an individual patient's future breast cancer risk from a mammogram so that screening intervals can be personalised rather than applied uniformly.

### Data and Infrastructure Requirements

Deploying an AI radiology system requires more than installing a software package; it depends on a supporting infrastructure of interoperable systems. Hospitals must maintain a PACS and Radiology Information System (RIS) capable of routing image data to the AI platform in real time, alongside sufficient computing capacity, whether on-premises graphics processing units or a secure cloud environment, to run inference without introducing delay into the clinical workflow. Data governance agreements must also be established to define how patient imaging data may be used, stored, and, where applicable, shared back with the vendor for ongoing model improvement, all while complying with health data privacy regulations such as HIPAA in the United States or equivalent frameworks elsewhere.

### Comparative Summary of AI Techniques

The techniques described above are not interchangeable; each is suited to a different stage of the diagnostic workflow, as summarised below.

| Technique | Primary Task | Example Use in Radiology |
|---|---|---|
| CNNs | Image classification and detection | Flagging suspected fractures or nodules on X-ray and CT |
| Segmentation Networks (e.g., U-Net) | Pixel/voxel-level delineation | Quantifying haemorrhage volume in Viz Subdural Plus |
| Foundation Models | Multi-condition detection from one input | Aidoc CARE model triaging 14 conditions per CT scan |
| NLP / Generative Models | Text generation and summarisation | Aidoc First Read drafting chest X-ray reports |
| Risk-Prediction Models | Longitudinal risk estimation | Mirai personalising mammogram screening intervals |

In practice, a single deployed platform often combines several of these techniques within one pipeline: a CNN-based classifier may perform initial triage, a segmentation network may then quantify the extent of an identified abnormality, and, increasingly, a language-model-based component may draft the accompanying report text, illustrating how modern radiology AI has moved from single-model tools toward integrated, multi-stage AI systems.

## Real-World Implementation and Case Studies

The table below summarises verified real-world deployments of AI radiology systems as of mid-2026, illustrating the scale and clinical focus areas of leading platforms.

| Platform | Primary Focus | Deployment Scale | Notable Milestone |
|---|---|---|---|
| Aidoc (aiOS / CARE) | Multi-condition CT triage | ~2,000 hospitals; 120M+ scans analysed | First foundation-model body CT triage, Jan 2026 |
| Viz.ai | Stroke (LVO) detection | 1,700+ hospitals; 50+ cleared algorithms | 44% faster stroke diagnosis (ISC 2025 trial) |
| Mirai (MIT/MGH) | Breast cancer risk prediction | Deployed at Mount Auburn Hospital, MA | Personalised mammogram screening intervals |
| GE HealthCare | Broad imaging AI portfolio | 120 radiology AI authorisations | Largest FDA authorisation count by vendor |

### Case Study: Stroke Detection with Viz.ai

Time is the single most critical factor in stroke outcomes, as delayed treatment directly increases the risk of permanent disability. Viz.ai's large-vessel-occlusion detection algorithm analyses CT angiography images and sends real-time smartphone alerts directly to on-call stroke teams, bypassing traditional sequential communication chains. In a multicentre study of 474 patients presented at the International Stroke Conference 2025, the system reduced stroke diagnosis time by 44 percent, cut time-to-treatment by 31 minutes on average, and reduced 90-day disability outcomes by 40 percent.

### Case Study: Multi-Condition Triage with Aidoc

Aidoc's platform is deployed across health systems including AdventHealth, WellSpan Health, Advocate Health, Mount Sinai, and Yale New Haven Health, and processes approximately 60 million patient scans annually. Its January 2026 clearance for a foundation-model-powered, multi-condition body CT triage tool marked a shift away from single-purpose algorithms toward a unified platform capable of flagging several critical findings from one scan and prioritising the radiologist's worklist accordingly.

### Case Study: Personalised Screening at Mount Auburn Hospital

Mount Auburn Hospital in Massachusetts has deployed the Mirai breast cancer risk model to tailor individual mammogram screening schedules, moving away from a uniform annual screening interval toward a risk-adjusted approach informed by each patient's AI-derived risk profile.

### Observations Across Implementations

Across these deployments, a consistent implementation pattern emerges. Successful adoption is associated with strong existing IT infrastructure, active clinician involvement during rollout, and an institutional commitment to ongoing local validation rather than a one-time installation. Academic medical centres such as Mount Sinai and Stanford, for example, have reported multi-disciplinary collaborations in which radiologists work directly alongside data scientists to fine-tune vendor algorithms on local patient data, often improving measured accuracy beyond the vendor's original published benchmarks. This suggests that the clinical value of an AI radiology tool is not fixed at the point of purchase but continues to depend on the receiving institution's willingness to invest in integration, training, and continuous performance monitoring after go-live.

## Benefits of AI in Radiology

- Faster Time-to-Diagnosis: Automated triage significantly reduces the time between image acquisition and specialist review for time-critical conditions such as stroke and pulmonary embolism.
- Reduced Clinician Workload: By reprioritising worklists and, increasingly, drafting preliminary reports, AI tools help address rising imaging volumes without a proportional increase in radiologist staffing.
- Improved Consistency: Deep learning models apply the same evaluation criteria to every scan, reducing the variability that can arise from fatigue or differences in individual reader experience.

- Personalised Screening: Risk-prediction models allow screening intervals to be tailored to individual patient risk rather than applying a uniform protocol to all patients.
- Scalability Across Health Systems: Once validated, a single platform can be deployed across many hospitals simultaneously, extending access to advanced diagnostic support in resource-constrained settings.
- Rapid Regulatory Progress: The FDA cleared 72 new AI-enabled medical devices in the final quarter of 2025 alone, with a median clearance time of 142 days, indicating a maturing and increasingly efficient regulatory pathway for validated tools.

Taken together, these benefits are most pronounced in conditions where clinical outcomes are highly sensitive to time, such as stroke, pulmonary embolism, and intracranial haemorrhage, where even a modest reduction in diagnosis-to-treatment time can materially change a patient's long-term functional outcome.

## Limitations and Challenges

- Low Real-World Utilisation: Despite the large number of cleared devices, only around 30 percent of radiologists currently use AI tools in routine clinical practice, revealing a significant gap between regulatory clearance and everyday clinical adoption.
- Narrow Regulatory Pathways: Around 95 to 97 percent of AI radiology clearances use the FDA's 510(k) pathway, which demonstrates substantial equivalence to an existing device rather than requiring the more rigorous clinical validation of the De Novo or Premarket Approval routes.
- Workflow Integration Burden: Successfully embedding an AI tool into a hospital's PACS and reading workflow requires substantial IT investment and change management, and integration difficulty is frequently underestimated during procurement.
- Limited Reimbursement: Insurance reimbursement for AI-assisted image interpretation remains rudimentary and, at present, exists in a meaningful way only for select cardiac imaging applications, limiting the financial incentive for widespread adoption.
- Population and Bias Risk: A model's performance can degrade significantly when applied to patient populations that differ from its training data, making local validation essential before deployment.
- Explainability and Trust: Deep learning models often function as "black boxes," and clinicians may be reluctant to act on an alert without a clear, interpretable rationale for the underlying prediction.
- Regulatory Evolution: Emerging frameworks such as the European Union's AI Act will require radiology AI to meet high-risk compliance obligations, including documentation of training data curation, bias assessment, and human oversight policies, adding further complexity for global vendors.

Many of these limitations are not purely technical but organisational, meaning that they can, in principle, be addressed through better procurement practices, clearer reimbursement policy, and stronger institutional commitment to post-deployment monitoring, rather than requiring fundamentally new AI architectures. This distinction is important for healthcare administrators evaluating AI investments: the central question is often not whether a tool is technically capable, but whether the receiving institution is organisationally prepared to integrate, monitor, and sustain it.

## Ethical, Regulatory, and Future Considerations

The continued expansion of AI in radiology raises several ethical and governance questions that extend beyond pure technical performance. Accountability for a missed or incorrect AI-assisted finding must be clearly defined between the device manufacturer, the deploying hospital, and the supervising clinician. Patient consent and transparency regarding the use of AI in their diagnostic pathway are increasingly discussed as necessary components of informed care. Data privacy is another central concern, given that training and continuous-learning pipelines rely on large volumes of sensitive patient imaging data drawn from multiple institutions.

Looking ahead, the field is moving toward multi-condition foundation models capable of analysing an entire scan for many pathologies simultaneously, rather than relying on a collection of narrow, single-purpose algorithms. The parallel emergence of generative AI tools for automated report drafting suggests that the next phase of radiology AI will extend beyond detection and triage into documentation and communication, provided that concerns regarding accuracy, liability, and clinician oversight are adequately addressed.

Workforce implications are also part of this future outlook. Contrary to early predictions that deep learning would substantially reduce demand for radiologists, real-world data instead shows radiologist salaries and demand at record highs, suggesting that AI is, in practice, changing the nature of the role rather than eliminating it. Routine, high-volume tasks such as initial triage and preliminary report drafting are increasingly automated, while radiologists are expected to spend more of their time on complex case interpretation, quality oversight of AI-flagged findings, and direct communication with referring clinicians.

## Critical Reflection: Does Clearance Equal Clinical Validation?

A recurring assumption in industry commentary is that a large and growing count of regulatory clearances is, by itself, evidence of clinical maturity. This report takes the position that clearance volume and clinical validation are related but distinct measures, and that treating them as interchangeable risks overstating how thoroughly these tools have actually been tested. A 2024 follow-up review of commercially available radiological AI products found that although the proportion of CE-certified products backed by at least one peer-reviewed study rose from 36 percent in 2020 to 66 percent by 2023, the proportion supported by higher-level evidence, that is, studies demonstrating an actual effect on clinical decision-making or patient outcomes rather than technical accuracy alone, remained largely unchanged at around 24 percent. In other words, most published evidence for commercial radiology AI still answers the narrower question of whether a model can detect a pattern in a test dataset, not the broader question of whether using it changes what a clinician does or how a patient fares.

A separate 2025 systematic review examining the generalisability of radiology AI models across different hospitals, scanners, and patient populations reached a related conclusion: external validation, meaning testing a model on data from an institution other than the one that trained it, remains uncommon, and performance can degrade meaningfully when a model encounters imaging equipment or patient demographics unlike its training data. Read together with the earlier observation that only around 30 percent of radiologists use AI tools in routine practice despite more than 1,500 cleared devices, a more critical reading emerges: the bottleneck limiting AI radiology adoption is not primarily a shortage of regulatory pathways or technical capability, but a shortage of the kind of outcome-level, externally validated evidence that would give clinicians firm grounds to trust an alert enough to act on it routinely. For a hospital evaluating a new AI tool, this suggests that the single most useful due-diligence question is not "has this been FDA cleared" but "has this been externally validated on a patient population similar to ours, and does that validation measure clinical outcomes rather than image-level accuracy alone."

## Conclusion

AI-based radiology systems represent one of the most mature and extensively deployed applications of Artificial Intelligence in healthcare today. Built primarily on convolutional neural networks, segmentation architectures, and increasingly on multimodal foundation models, these systems are already embedded in the diagnostic workflows of thousands of hospitals worldwide, delivering measurable improvements in the speed and consistency of time-critical diagnoses such as stroke and pulmonary embolism. At the same time, the gap between regulatory clearance and real-world clinical adoption, combined with unresolved challenges around reimbursement, explainability, and workflow integration, shows that this technology remains in an active and evolving stage of maturity rather than a finished solution. As foundation models, generative reporting tools, and evolving regulatory frameworks continue to develop, AI radiology systems are likely to move from a supporting triage role toward a more deeply integrated position within the broader diagnostic and reporting workflow of modern healthcare.

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**Rubrics for Evaluation**

| S.No | Criteria | Weightage (%) | **Excellent** (100—90%) | **Good** (89—75%) | **Satisfactory (74-60%)** | **Improvement (** |
|---|---|---|---|---|---|---|
| 1. | Identification of Latest AI Application | 10% | Selects a highly relevant, emerging AI application with strong justification. | Selects a recent and relevant AI application. | Selects a suitable application with limited justification. | Application is outdated or irrelevant. |
| 2. | Problem Analysis and Domain Understanding | 15% | Clearly explains the real-world problem, domain context, and need for AI with critical analysis. | Good explanation with adequate analysis. | Basic explanation with limited analysis. | Problem statement is unclear or incomplete. |
| 3. | AI Technologies and Working Mechanism | 20% | Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth. | Good explanation with minor omissions. | Covers only basic AI concepts. | AI technologies are poorly explained or incorrect. |
| 4. | Research Quality and Literature Review | 10% | Uses recent, credible sources with proper citations and excellent synthesis. | Good quality references with appropriate citations. | Limited references with basic discussion. | Few or unreliable references; poor citation. |
| 5. | Real-World Implementation and Case Analysis | **10$^{0}$/0** | Thoroughly analyzes industrial implementation with appropriate examples. | Good implementation analysis. | Basic implementation discussion. | Minimal or no real-world analysis. |
| 6. | Impact, Challenges, and Future Scope | 15% | Critically evaluates benefits, limitations, ethical issues, and future trends. | Good evaluation with minor gaps. | Basic **discussion** of impact and future scope. | Limited evaluation or missing sections. |
| 7. | Originality and Critical Thinking | 10$^{0}$/0 | Demonstrates independent research, innovation, and critical evaluation. | Good analytical thinking. | Some originality with limited analysis. | Mostly descriptive or copied content. |
| 8. | Organization, Presentation, and Documentation | 10$^{0}$ 0 | Exceptionally organized, professional presentation, error-free, and properly referenced. | Well organized with minor issues. | Adequately organized with some errors. | Poor organization and documentation. |